

Spring Boot + Thymeleaf 完整教學指南

顧客管理系統實作

📖 學習目標

通過本教學，您將學會：

- Spring Boot MVC 架構設計
- Thymeleaf 模板引擎的使用
- 表單處理與資料綁定
- RESTful 控制器設計
- 前後端資料傳遞最佳實務

📁 專案結構

```
src/main/java/
├── com.example.sbweb1113/
│   ├── controller/
│   │   └── ViewController.java
│   ├── model/
│   │   └── Customer.java
│   └── SbWeb1113Application.java
src/main/resources/
├── templates/
│   ├── index.html
│   ├── customerlist.html
│   ├── viewcustomer.html
│   └── customerupdate.html
└── application.properties
```

🔗 Maven 依賴設定

```
<dependencies>
  <!-- Spring Boot Web Starter -->
  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-web</artifactId>
  </dependency>

  <!-- Thymeleaf Template Engine -->
  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-thymeleaf</artifactId>
  </dependency>

  <!-- Spring Boot DevTools (開發用) -->
```

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-devtools</artifactId>
  <scope>runtime</scope>
  <optional>true</optional>
</dependency>
</dependencies>
```

第一部分：資料模型設計

Customer.java - 顧客實體類別

```
package com.example.sbweb1113.model;

import javax.validation.constraints.*;

/**
 * 顧客實體類別
 * 包含基本的顧客信息和驗證規則
 */
public class Customer {

    @NotBlank(message = "姓名不能為空")
    @Size(min = 2, max = 50, message = "姓名長度必須在2-50字符之間")
    private String name;

    @NotBlank(message = "地址不能為空")
    @Size(max = 200, message = "地址長度不能超過200字符")
    private String address;

    @Pattern(regexp = "^[0-9]+$", message = "重量必須是數字")
    @NotBlank(message = "重量不能為空")
    private String weight;

    // 預設建構子
    public Customer() {}

    // 完整建構子
    public Customer(String name, String address, String weight) {
        this.name = name;
        this.address = address;
        this.weight = weight;
    }

    // Getter 和 Setter 方法
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }

    public String getAddress() { return address; }
    public void setAddress(String address) { this.address = address; }
```

```
public String getWeight() { return weight; }
public void setWeight(String weight) { this.weight = weight; }

@Override
public String toString() {
    return "Customer{name='" + name + "', address='" + address + "', weight='"
+ weight + "'}";
}
}
```

第二部分：控制器設計

ViewController.java - MVC 控制器

```
package com.example.sbweb1113.controller;

import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.ModelAttribute;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestParam;
import org.springframework.web.servlet.ModelAndView;

import com.example.sbweb1113.model.Customer;

import java.util.*;

@Controller
public class ViewController {
    /**
     * 🏠 首頁 - 基本字串傳遞示範
     * URL: /nameview
     * 功能：展示如何將簡單字串傳遞到 Thymeleaf 模板
     */
    @GetMapping(value="/nameview")
    public String home(Model model) {
        model.addAttribute("username", "ben");
        return "index.html";
    }

    /**
     * 👤 顧客檢視 - Map 資料傳遞示範
     * URL: /customerview
     * 功能：使用 Map 將顧客資料傳遞到模板
     */
    @GetMapping(value="/customerview")
    public String vcustomer(Model model) {
        Map<String,String> data=new HashMap<>();
```

```
        data.put("name", "Henrry");
        data.put("address", "Kung Yuan Road");
        data.put("weight", "60");
        model.addAttribute("customer", data);
        return "viewcustomer.html";
    }

    /**
     * + 新增顧客 - 表單資料處理
     * URL: /customerCreate (POST)
     * 功能：處理表單提交的顧客資料，使用 @ModelAttribute 綁定
     */
    @PostMapping(value="/customerCreate")
    public String addCustomer(@ModelAttribute Customer data, Model model) {
        model.addAttribute("customer", data);
        return "viewcustomer.html";
    }

    /**
     * 📄 JSON 顧客資料處理
     * URL: /jsoncustomer (POST)
     * 功能：處理 JSON 格式的顧客資料，使用 @RequestBody 綁定
     */
    @PostMapping(value="/jsoncustomer")
    public String addJsonCustomer(@RequestBody Customer data, Model model) {
        model.addAttribute("customer", data);
        return "viewcustomer.html";
    }

    /**
     * 🔄 顧客更新頁面 - URL 參數處理
     * URL: /customerUpdate?id=name
     * 功能：根據 URL 參數載入顧客資料進行編輯
     */
    @GetMapping(value="/customerUpdate")
    public String customerUpdate(@RequestParam("id")String name,Model model) {
        Customer data=new Customer();
        data.setName(name);
        data.setAddress("NA");
        data.setWeight("0");
        model.addAttribute("customer", data);
        return "customerupdate.html";
    }

    /**
     * 🔄 顧客更新頁面 - 預設資料示範
     * URL: /customerupdate
     * 功能：顯示編輯表單，預載入範例顧客資料
     */
    @GetMapping(value="/customerupdate")
    public String v2customer(Model model) {
        Customer data=new Customer("Henrry","Fu Yuan Road","50");
        model.addAttribute("customer", data);
        return "customerupdate.html";
    }
}
```

```
}

/**
 * 🔄 顧客更新頁面 - ModelAndView 示範
 * URL: /customerupdate2
 * 功能：使用 ModelAndView 方式返回視圖和資料
 */
@GetMapping(value="/customerupdate2")
public ModelAndView v3customer() {
    ModelAndView mv=new ModelAndView("customerupdate");
    Customer data=new Customer("Danis","Fu Yuan Road","55");
    mv.addObject("customer", data);
    return mv;
}

/**
 * 📋 顧客清單 - 集合資料處理
 * URL: /customerlist
 * 功能：顯示所有顧客的清單，展示 Thymeleaf 迴圈處理
 */
@GetMapping(value="/customerlist")
public String listcustomer(Model model) {
    List<Customer> data=List.of(
        new Customer("Alan","Fu Yuan Road","50"),
        new Customer("White","Lily Road","55"),
        new Customer("Rose","Flower Road","53"));

    model.addAttribute("customers", data);
    return "customerlist.html";
}

/**
 * 📝 處理顧客表單提交
 * URL: /customerreceive (POST)
 * 功能：接收更新後的顧客資料並重導向到列表頁面
 */
@PostMapping(value="/customerreceive")
public String receiveCustomer(@ModelAttribute Customer customer, Model model)
{
    // 在實際應用中，這裡會保存到資料庫
    System.out.println("收到顧客資料: " + customer);

    // 重導向到顧客列表頁面
    return "redirect:/customerlist";
}

/**
 * ✕ 刪除顧客 - URL 參數處理
 * URL: /customerDelete?id=name
 * 功能：根據顧客姓名刪除顧客 (示範用)
 */
@GetMapping(value="/customerDelete")
public String deleteCustomer(@RequestParam("id") String name, Model model) {
    // 在實際應用中，這裡會從資料庫刪除
}
```

```
        System.out.println("刪除顧客: " + name);

        // 重導向到顧客列表頁面
        return "redirect:/customerlist";
    }
}
---
```

🧠 第三部分：Thymeleaf 模板設計

📖 Thymeleaf 語法速覽

功能	語法	說明
文字輸出	<code>`th:text="\${variable}"`</code>	顯示變數內容
HTML輸出	<code>`th:utext="\${variable}"`</code>	顯示HTML內容
屬性設定	<code>`th:attr="href=\${url}"`</code>	設定屬性值
條件判斷	<code>`th:if="\${condition}"`</code>	條件顯示
迴圈處理	<code>`th:each="item : \${list}"`</code>	遍歷集合
表單綁定	<code>`th:object="\${object}"`</code>	綁定表單物件
欄位綁定	<code>`th:field="*{property}"`</code>	綁定物件屬性
URL產生	<code>`th:href="@{/path(param=\${value})}"`</code>	生成連結

🏠 index.html - 基本文字顯示

```
```.html
<!DOCTYPE html>
<html lang="zh-TW" xmlns:th="http://www.thymeleaf.org">
<head>
 <meta charset="UTF-8"/>
 <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
 <title>Spring Boot + Thymeleaf 示範</title>
 <style>
 body {
 font-family: 'Microsoft JhengHei', Arial, sans-serif;
 margin: 40px;
 background-color: #f5f5f5;
 }
 .container {
 max-width: 800px;
 margin: 0 auto;
 background: white;
 padding: 30px;
 border-radius: 10px;
 box-shadow: 0 2px 10px rgba(0,0,0,0.1);
 }
 h1 { color: #2c3e50; }
 .welcome {
 font-size: 18px;
 color: #27ae60;
 margin: 20px 0;
 }
 .nav-links a {
 display: inline-block;
```

```

 margin: 10px 15px 10px 0;
 padding: 10px 20px;
 background: #3498db;
 color: white;
 text-decoration: none;
 border-radius: 5px;
 }
 .nav-links a:hover { background: #2980b9; }
</style>
</head>
<body>
 <div class="container">
 <h1>👋 歡迎使用 Spring Boot + Thymeleaf</h1>

 <!-- Thymeleaf 文字輸出示範 -->
 <div class="welcome">
 🤖 您好，訪客！
 </div>

 <!-- 導航連結 -->
 <div class="nav-links">
 <h3>📖 教學示範頁面：</h3>
 <a th:href="@{/customerview}">🔍 檢視顧客資料
 <a th:href="@{/customerlist}">📋 顧客清單
 <a th:href="@{/customerupdate}">✎ 編輯顧客
 </div>

 <!-- 新增顧客表單 -->
 <div style="margin-top: 30px; padding: 20px; background: #ecf0f1; border-
radius: 5px;">
 <h3>➕ 新增顧客</h3>
 <form th:action="@{/customerCreate}" method="post" style="display:
grid; gap: 10px; max-width: 400px;">
 <div>
 <label>👤 姓名：</label>
 <input type="text" name="name" required style="width: 200px;
padding: 5px;"/>
 </div>
 <div>
 <label>🏠 地址：</label>
 <input type="text" name="address" required style="width:
200px; padding: 5px;"/>
 </div>
 <div>
 <label>🏋️ 重量：</label>
 <input type="number" name="weight" required style="width:
200px; padding: 5px;"/>
 </div>
 <div>
 <button type="submit" style="width: 100px; padding: 8px;
background: #27ae60; color: white; border: none; border-radius: 3px;">
 新增
 </button>
 </div>
 </form>
 </div>
 </div>

```

```
 </form>
 </div>
 </div>
 </body>
</html>
```

## 📄 customerlist.html - 清單與迴圈處理

```
<!DOCTYPE html>
<html lang="zh-TW" xmlns:th="http://www.thymeleaf.org">
<head>
 <meta charset="UTF-8"/>
 <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
 <title>顧客清單管理</title>
 <style>
 body {
 font-family: 'Microsoft JhengHei', Arial, sans-serif;
 margin: 20px;
 background-color: #f8f9fa;
 }
 .container {
 max-width: 1000px;
 margin: 0 auto;
 background: white;
 padding: 30px;
 border-radius: 10px;
 box-shadow: 0 2px 15px rgba(0,0,0,0.1);
 }
 h1 {
 color: #2c3e50;
 border-bottom: 3px solid #3498db;
 padding-bottom: 10px;
 }
 .header-actions {
 margin: 20px 0;
 text-align: right;
 }
 .btn {
 display: inline-block;
 padding: 10px 20px;
 color: white;
 text-decoration: none;
 border-radius: 5px;
 margin: 0 5px;
 }
 .btn-primary { background-color: #3498db; }
 .btn-primary:hover { background-color: #2980b9; }
 .btn-success { background-color: #27ae60; }
 .btn-success:hover { background-color: #229954; }
 .btn-warning { background-color: #f39c12; }
 .btn-warning:hover { background-color: #e67e22; }
```



```

.btn-danger { background-color: #e74c3c; }
.btn-danger:hover { background-color: #c0392b; }

table {
 width: 100%;
 border-collapse: collapse;
 margin-top: 20px;
 box-shadow: 0 1px 5px rgba(0,0,0,0.1);
}
th {
 background-color: #34495e;
 color: white;
 padding: 15px;
 text-align: left;
 font-weight: bold;
}
td {
 padding: 12px 15px;
 border-bottom: 1px solid #ddd;
}
tr:nth-child(even) {
 background-color: #f8f9fa;
}
tr:hover {
 background-color: #e8f4fd;
}
.actions {
 white-space: nowrap;
}
.actions a {
 font-size: 12px;
 padding: 5px 10px;
 margin: 0 2px;
}
.empty-message {
 text-align: center;
 color: #7f8c8d;
 font-style: italic;
 padding: 40px;
}
</style>
</head>
<body>
 <div class="container">
 <h1>📧 顧客清單管理</h1>

 <div class="header-actions">
 <a th:href="@{/}" class="btn btn-primary">🏠 回到首頁
 <a th:href="@{/customerupdate}" class="btn btn-success">+ 新增顧客

 </div>

 <!-- Thymeleaf 條件判斷：檢查是否有顧客資料 -->
 <div th:if="${customers != null and #lists.size(customers) > 0}">

```

```

<p> 總計：0 位顧客</p>

<table>
 <thead>
 <tr>
 <th> 姓名</th>
 <th> 地址</th>
 <th> 重量</th>
 <th> 操作</th>
 </tr>
 </thead>
 <tbody>
 <!-- Thymeleaf 迴圈：遍歷顧客清單 -->
 <tr th:each="customer, iterStat : ${customers}">
 <td>
 姓名
 <!-- 顯示索引號碼 -->
 <small th:text="'(' + ${iterStat.count} + ')">
</small>

 </td>
 <td th:text="${customer.address}">地址</td>
 <td>
 重量
 <small>kg</small>
 </td>
 <td class="actions">
 <!-- Thymeleaf URL 產生：帶參數的連結 -->
 <a th:href="@{/customerUpdate(id=${customer.name})}"
 class="btn btn-warning"> 修改
 <a th:href="@{/customerDelete(id=${customer.name})}"
 class="btn btn-danger"
 onclick="return confirm('確定要刪除這位顧客嗎?')">
 刪除

 </td>
 </tr>
 </tbody>
</table>
</div>

<!-- Thymeleaf 條件判斷：沒有資料時的顯示 -->
<div th:if="${customers == null or #lists.size(customers) == 0}"
class="empty-message">
 <h3> 目前沒有顧客資料</h3>
 <p>點擊上方的「新增顧客」按鈕來新增第一位顧客吧！</p>
</div>
</div>
</body>
</html>

```

```
<!DOCTYPE html>
<html lang="zh-TW" xmlns:th="http://www.thymeleaf.org">
<head>
 <meta charset="UTF-8"/>
 <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
 <title>顧客資料檢視</title>
 <style>
 body {
 font-family: 'Microsoft JhengHei', Arial, sans-serif;
 margin: 20px;
 background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
 min-height: 100vh;
 }
 .container {
 max-width: 600px;
 margin: 50px auto;
 background: white;
 padding: 40px;
 border-radius: 15px;
 box-shadow: 0 10px 30px rgba(0,0,0,0.2);
 }
 h1 {
 color: #2c3e50;
 text-align: center;
 margin-bottom: 30px;
 }
 .customer-card {
 background: #f8f9fa;
 border-left: 5px solid #3498db;
 padding: 25px;
 border-radius: 8px;
 margin: 20px 0;
 }
 .customer-field {
 display: flex;
 align-items: center;
 margin: 15px 0;
 font-size: 18px;
 }
 .field-label {
 min-width: 80px;
 font-weight: bold;
 color: #34495e;
 }
 .field-value {
 color: #27ae60;
 font-weight: 500;
 margin-left: 10px;
 }
 .actions {
 text-align: center;
 margin-top: 30px;
 }
 </style>
</head>
<body>
 <div class="container">
 <h1>顧客資料檢視</h1>
 <div class="customer-card">
 <div class="customer-field">
 <div class="field-label">姓名</div>
 <div class="field-value">張三</div>
 </div>
 <div class="customer-field">
 <div class="field-label">電話</div>
 <div class="field-value">0912345678</div>
 </div>
 <div class="customer-field">
 <div class="field-label">地址</div>
 <div class="field-value">台北市中山區</div>
 </div>
 <div class="actions">
 <button>新增</button>
 <button>刪除</button>
 </div>
 </div>
 </div>
</body>
</html>
```

```

 .btn {
 display: inline-block;
 padding: 12px 25px;
 margin: 0 10px;
 color: white;
 text-decoration: none;
 border-radius: 6px;
 font-weight: bold;
 transition: all 0.3s;
 }
 .btn-primary { background-color: #3498db; }
 .btn-primary:hover { background-color: #2980b9; transform:
translateY(-2px); }
 .btn-secondary { background-color: #95a5a6; }
 .btn-secondary:hover { background-color: #7f8c8d; transform:
translateY(-2px); }
 .btn-warning { background-color: #f39c12; }
 .btn-warning:hover { background-color: #e67e22; transform:
translateY(-2px); }
 </style>
</head>
<body>
 <div class="container">
 <h1> 顧客資料詳情</h1>

 <div class="customer-card">
 <div class="customer-field">
 姓名:
 顧客姓名

 </div>

 <div class="customer-field">
 地址:
 顧客地址

 </div>

 <div class="customer-field">
 重量:

 重量 kg

 </div>

 <!-- Thymeleaf 表達式：計算BMI（假設身高170cm） -->
 <div class="customer-field" th:if="${customer.weight != null and
customer.weight != '0'}">
 BMI:
 <span class="field-value"
th:text="${#numbers.formatDecimal(T(Double).parseDouble(customer.weight) / (1.7 *
1.7), 1, 2)}">
 計算中

 </div>
 </div>
 </div>

```

```


 <small>(假設身高170cm)</small>
 </div>
</div>

<div class="actions">
 <a th:href="@{/}" class="btn btn-primary">🏠 回首頁
 <a th:href="@{/customerlist}" class="btn btn-secondary">📋 顧客清單

 <a th:href="@{/customerUpdate(id=${customer.name})}" class="btn btn-
warning">✎ 編輯資料
</div>
</div>
</body>
</html>

```

## ✎ customerupdate.html - 表單處理

```

<!DOCTYPE html>
<html lang="zh-TW" xmlns:th="http://www.thymeleaf.org">
<head>
 <meta charset="UTF-8"/>
 <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
 <title>編輯顧客資料</title>
 <style>
 body {
 font-family: 'Microsoft JhengHei', Arial, sans-serif;
 margin: 20px;
 background: linear-gradient(135deg, #a8edea 0%, #fed6e3 100%);
 min-height: 100vh;
 }
 .container {
 max-width: 500px;
 margin: 50px auto;
 background: white;
 padding: 40px;
 border-radius: 15px;
 box-shadow: 0 8px 25px rgba(0,0,0,0.1);
 }
 h1 {
 color: #2c3e50;
 text-align: center;
 margin-bottom: 30px;
 border-bottom: 3px solid #e74c3c;
 padding-bottom: 10px;
 }
 .form-group {
 margin-bottom: 20px;
 }
 label {
 display: block;

```

```
 margin-bottom: 5px;
 font-weight: bold;
 color: #34495e;
 font-size: 16px;
 }
 input[type="text"], input[type="number"] {
 width: 100%;
 padding: 12px 15px;
 border: 2px solid #bdc3c7;
 border-radius: 8px;
 font-size: 16px;
 transition: border-color 0.3s;
 box-sizing: border-box;
 }
 input[type="text"]:focus, input[type="number"]:focus {
 outline: none;
 border-color: #3498db;
 box-shadow: 0 0 5px rgba(52, 152, 219, 0.3);
 }
 input[readonly] {
 background-color: #f1f2f6;
 color: #7f8c8d;
 cursor: not-allowed;
 }
 .form-actions {
 text-align: center;
 margin-top: 30px;
 padding-top: 20px;
 border-top: 1px solid #ecf0f1;
 }
 .btn {
 display: inline-block;
 padding: 12px 30px;
 margin: 0 10px;
 border: none;
 border-radius: 6px;
 font-size: 16px;
 font-weight: bold;
 text-decoration: none;
 cursor: pointer;
 transition: all 0.3s;
 }
 .btn-primary {
 background-color: #27ae60;
 color: white;
 }
 .btn-primary:hover {
 background-color: #229954;
 transform: translateY(-2px);
 }
 .btn-secondary {
 background-color: #95a5a6;
 color: white;
 }
}
```

```

 .btn-secondary:hover {
 background-color: #7f8c8d;
 transform: translateY(-2px);
 }
 .required {
 color: #e74c3c;
 }
 .form-note {
 background-color: #f8f9fa;
 padding: 15px;
 border-radius: 5px;
 border-left: 4px solid #17a2b8;
 margin-bottom: 20px;
 color: #495057;
 }
 }
</style>
</head>
<body>
 <div class="container">
 <h1>✎ 編輯顧客資料</h1>

 <div class="form-note">
 💡 提示 :帶有 * 的欄位為
 必填項目
 </div>

 <!-- Thymeleaf 表單物件綁定 : th:object="${customer}" -->
 <form th:action="@{/customerreceive}" th:object="${customer}"
 method="post">

 <div class="form-group">
 <label for="name">👤 顧客姓名 *
 </label>

 <!-- Thymeleaf 欄位綁定 : th:field="*{name}" -->
 <input type="text"
 id="name"
 th:field="*{name}"
 required
 placeholder="請輸入顧客姓名"
 maxlength="50"/>

 </div>

 <div class="form-group">
 <label for="address">🏠 地址 *
 </label>

 <!-- th:field 會自動處理 name, id, value 屬性 -->
 <input type="text"
 id="address"
 th:field="*{address}"
 required
 placeholder="請輸入詳細地址"
 maxlength="200"/>

 </div>

```

```

 <div class="form-group">
 <label for="weight">🐾 重量 (公斤) *
 </label>

 <input type="number"
 id="weight"
 th:field="*{weight}"
 required
 min="1"
 max="500"
 step="0.1"
 placeholder="請輸入重量"/>

 </div>

 <div class="form-actions">
 <button type="submit" class="btn btn-primary">💾 儲存變更</button>
 <a th:href="@{/customerlist}" class="btn btn-secondary">✕ 取消

 </div>
 </form>

 <!-- 額外的導航連結 -->
 <div style="text-align: center; margin-top: 20px; padding-top: 15px;
border-top: 1px solid #ecf0f1;">
 <a th:href="@{/}" style="color: #3498db; text-decoration: none;">🏠 回
到首頁
 </div>
</div>
</body>
</html>

```

## ⚙️ 第四部分：專案配置與進階功能

### 📄 完整的 application.properties 配置

```

應用程式基本配置
spring.application.name=spring-boot-thymeleaf-demo
server.port=8080
server.servlet.context-path=/sbweb1113

Thymeleaf 模板引擎配置
spring.thymeleaf.prefix=classpath:/templates/
spring.thymeleaf.suffix=.html
spring.thymeleaf.mode=HTML
spring.thymeleaf.encoding=UTF-8
spring.thymeleaf.servlet.content-type=text/html
spring.thymeleaf.cache=false
spring.thymeleaf.check-template=true

開發環境配置
spring.devtools.restart.enabled=true

```



```
spring.devtools.livereload.enabled=true

靜態資源配置
spring.mvc.static-path-pattern=/static/**
spring.web.resources.static-locations=classpath:/static/

日誌配置
logging.level.org.springframework.web=INFO
logging.level.com.example.sbweb1113=DEBUG
```

## 增強版 Customer 實體類別

```
package com.example.sbweb1113.model;

import javax.validation.constraints.*;
import java.time.LocalDateTime;

/**
 * 增強版顧客實體類別
 * 包含驗證規則和審計欄位
 */
public class Customer {

 @NotBlank(message = "姓名不能為空")
 @Size(min = 2, max = 50, message = "姓名長度必須在2-50字符之間")
 private String name;

 @NotBlank(message = "地址不能為空")
 @Size(max = 200, message = "地址長度不能超過200字符")
 private String address;

 @Pattern(regexp = "^[0-9]+(\\.[0-9]+)?$", message = "重量必須是有效數字")
 @NotBlank(message = "重量不能為空")
 private String weight;

 @Email(message = "請輸入有效的電子郵件格式")
 private String email;

 private String phone;
 private LocalDateTime createdAt;
 private LocalDateTime updatedAt;

 // 建構子
 public Customer() {
 this.createdAt = LocalDateTime.now();
 this.updatedAt = LocalDateTime.now();
 }

 public Customer(String name, String address, String weight) {
 this();
 this.name = name;
 }
}
```

```
 this.address = address;
 this.weight = weight;
 }

 public Customer(String name, String address, String weight, String email,
String phone) {
 this(name, address, weight);
 this.email = email;
 this.phone = phone;
 }

 // Getter 和 Setter 方法
 public String getName() { return name; }
 public void setName(String name) {
 this.name = name;
 this.updatedAt = LocalDateTime.now();
 }

 public String getAddress() { return address; }
 public void setAddress(String address) {
 this.address = address;
 this.updatedAt = LocalDateTime.now();
 }

 public String getWeight() { return weight; }
 public void setWeight(String weight) {
 this.weight = weight;
 this.updatedAt = LocalDateTime.now();
 }

 public String getEmail() { return email; }
 public void setEmail(String email) {
 this.email = email;
 this.updatedAt = LocalDateTime.now();
 }

 public String getPhone() { return phone; }
 public void setPhone(String phone) {
 this.phone = phone;
 this.updatedAt = LocalDateTime.now();
 }

 public LocalDateTime getCreatedAt() { return createdAt; }
 public void setCreatedAt(LocalDateTime createdAt) { this.createdAt =
createdAt; }

 public LocalDateTime getUpdatedAt() { return updatedAt; }
 public void setUpdatedAt(LocalDateTime updatedAt) { this.updatedAt =
updatedAt; }

 // 輔助方法
 public double getWeightAsDouble() {
 try {
 return Double.parseDouble(this.weight);
 }
```

```

 } catch (NumberFormatException e) {
 return 0.0;
 }
 }

 public double calculateBMI(double heightInMeters) {
 return getWeightAsDouble() / (heightInMeters * heightInMeters);
 }

 public String getWeightCategory() {
 double w = getWeightAsDouble();
 if (w < 60) return "輕量級";
 if (w > 75) return "重量級";
 return "標準級";
 }

 @Override
 public String toString() {
 return String.format("Customer{name='%s', address='%s', weight='%s', email='%s'}",
 name, address, weight, email);
 }
}

```

## 進階控制器功能

```

package com.example.sbweb1113.controller;

import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.validation.BindingResult;
import org.springframework.web.bind.annotation.*;
import org.springframework.web.servlet.mvc.support.RedirectAttributes;
import com.example.sbweb1113.model.Customer;

import javax.validation.Valid;
import java.util.*;
import java.util.stream.Collectors;

@Controller
public class AdvancedViewController {

 // 模擬資料庫
 private static List<Customer> customerDatabase = new ArrayList<>
(Arrays.asList(
 new Customer("Alice Johnson", "台北市信義區忠孝東路123號", "55",
"alice@email.com", "0912-345-678"),
 new Customer("Bob Smith", "新北市板橋區文化路456號", "70", "bob@email.com",
"0923-456-789"),
 new Customer("Carol Davis", "桃園市中壢區中正路789號", "48",
"carol@email.com", "0934-567-890"),

```

```
 new Customer("David Brown", "台中市西屯區台灣大道321號", "82",
 "david@email.com", "0945-678-901"),
 new Customer("Emily Wilson", "高雄市前金區中正四路654號", "52",
 "emily@email.com", "0956-789-012")
));

/**
 * 🏠 改良版首頁
 */
@GetMapping("/")
public String homepage(Model model) {
 model.addAttribute("customerCount", customerDatabase.size());
 model.addAttribute("averageWeight", calculateAverageWeight());
 model.addAttribute("recentCustomers", getRecentCustomers(3));
 model.addAttribute("systemStats", getSystemStats());
 return "homepage";
}

/**
 * 📊 顧客統計儀表板
 */
@GetMapping("/customerStats")
public String customerStatistics(Model model) {
 Map<String, Object> stats = generateStatistics();
 model.addAttribute("statistics", stats);
 model.addAttribute("customers", customerDatabase);
 model.addAttribute("chartData", generateChartData());
 return "customerstats";
}

/**
 * 🔍 進階搜尋功能
 */
@GetMapping("/customerAdvancedSearch")
public String advancedSearch(
 @RequestParam(required = false) String name,
 @RequestParam(required = false) String city,
 @RequestParam(required = false) String minWeight,
 @RequestParam(required = false) String maxWeight,
 @RequestParam(required = false) String email,
 Model model) {

 List<Customer> filteredCustomers = performAdvancedSearch(name, city,
 minWeight, maxWeight, email);

 model.addAttribute("customers", filteredCustomers);
 model.addAttribute("searchParams", buildSearchParams(name, city,
 minWeight, maxWeight, email));
 model.addAttribute("resultCount", filteredCustomers.size());
 model.addAttribute("totalCount", customerDatabase.size());
 model.addAttribute("searchSummary", generateSearchSummary(name, city,
 minWeight, maxWeight, email));

 return "customeradvancedsearch";
}
```

```
}

/**
 * 📄 完整的顧客表單處理
 */
@GetMapping("/customerForm")
public String showCustomerForm(@RequestParam(required = false) Integer index,
Model model) {
 Customer customer;
 boolean isEdit = false;

 if (index != null && index >= 0 && index < customerDatabase.size()) {
 customer = customerDatabase.get(index);
 isEdit = true;
 model.addAttribute("customerIndex", index);
 } else {
 customer = new Customer();
 }

 model.addAttribute("customer", customer);
 model.addAttribute("isEdit", isEdit);
 model.addAttribute("cities", getCityOptions());

 return "customerform";
}

/**
 * 📄 提交顧客表單
 */
@PostMapping("/customerSubmit")
public String submitCustomer(
 @Valid @ModelAttribute Customer customer,
 BindingResult bindingResult,
 @RequestParam(required = false) Integer customerIndex,
 Model model,
 RedirectAttributes redirectAttributes) {

 // 自訂驗證
 if (!bindingResult.hasErrors()) {
 performCustomValidation(customer, bindingResult, customerIndex);
 }

 if (bindingResult.hasErrors()) {
 model.addAttribute("customer", customer);
 model.addAttribute("isEdit", customerIndex != null);
 model.addAttribute("customerIndex", customerIndex);
 model.addAttribute("cities", getCityOptions());
 model.addAttribute("validationErrors", bindingResult.getAllErrors());
 return "customerform";
 }

 // 保存或更新顧客
 String message;
 if (customerIndex != null && customerIndex >= 0 && customerIndex <
```

```
customerDatabase.size()) {
 customerDatabase.set(customerIndex, customer);
 message = "✅ 顧客「" + customer.getName() + "」資料更新成功！";
} else {
 customerDatabase.add(customer);
 message = "✅ 顧客「" + customer.getName() + "」新增成功！";
}

redirectAttributes.addFlashAttribute("successMessage", message);
return "redirect:/customerlist";
}

/**
 * 🗑️ 刪除顧客 (POST 方式更安全)
 */
@PostMapping("/customerDelete/{index}")
public String deleteCustomer(
 @PathVariable int index,
 RedirectAttributes redirectAttributes) {

 if (index < 0 || index >= customerDatabase.size()) {
 redirectAttributes.addFlashAttribute("errorMessage", "❌ 找不到要刪除的顧客！");
 return "redirect:/customerlist";
 }

 Customer deletedCustomer = customerDatabase.remove(index);
 redirectAttributes.addFlashAttribute("successMessage",
 "🗑️ 顧客「" + deletedCustomer.getName() + "」已成功刪除！");

 return "redirect:/customerlist";
}

/**
 * 🏠 AJAX API 端點
 */
@GetMapping("/api/customers")
@ResponseBody
public Map<String, Object> getCustomersApi(
 @RequestParam(defaultValue = "0") int page,
 @RequestParam(defaultValue = "10") int size) {

 int start = page * size;
 int end = Math.min(start + size, customerDatabase.size());

 List<Customer> pageCustomers = customerDatabase.subList(start, end);

 Map<String, Object> response = new HashMap<>();
 response.put("customers", pageCustomers);
 response.put("totalElements", customerDatabase.size());
 response.put("totalPages", (int) Math.ceil((double)
customerDatabase.size() / size));
 response.put("currentPage", page);
 response.put("pageSize", size);
}
```

```
 response.put("timestamp", new Date());

 return response;
 }

 /**
 *  統計 API
 */
 @GetMapping("/api/statistics")
 @ResponseBody
 public Map<String, Object> getStatisticsApi() {
 return generateStatistics();
 }

 // === 輔助方法 ===

 private List<Customer> performAdvancedSearch(String name, String city, String
minWeight, String maxWeight, String email) {
 return customerDatabase.stream()
 .filter(customer -> {
 boolean matches = true;

 if (name != null && !name.trim().isEmpty()) {
 matches &=
customer.getName().toLowerCase().contains(name.toLowerCase());
 }

 if (city != null && !city.trim().isEmpty()) {
 matches &=
customer.getAddress().toLowerCase().contains(city.toLowerCase());
 }

 if (email != null && !email.trim().isEmpty()) {
 matches &= customer.getEmail() != null &&
customer.getEmail().toLowerCase().contains(email.toLowerCase());
 }

 if (minWeight != null && !minWeight.trim().isEmpty()) {
 try {
 double min = Double.parseDouble(minWeight);
 matches &= customer.getWeightAsDouble() >= min;
 } catch (NumberFormatException ignored) {}
 }

 if (maxWeight != null && !maxWeight.trim().isEmpty()) {
 try {
 double max = Double.parseDouble(maxWeight);
 matches &= customer.getWeightAsDouble() <= max;
 } catch (NumberFormatException ignored) {}
 }

 return matches;
 })
 })
```

```
 .collect(Collectors.toList());
 }

 private Map<String, String> buildSearchParams(String name, String city, String
minWeight, String maxWeight, String email) {
 Map<String, String> params = new HashMap<>();
 params.put("name", name != null ? name : "");
 params.put("city", city != null ? city : "");
 params.put("minWeight", minWeight != null ? minWeight : "");
 params.put("maxWeight", maxWeight != null ? maxWeight : "");
 params.put("email", email != null ? email : "");
 return params;
 }

 private String generateSearchSummary(String name, String city, String
minWeight, String maxWeight, String email) {
 List<String> conditions = new ArrayList<>();
 if (name != null && !name.trim().isEmpty()) conditions.add("姓名包含「" +
name + "」");
 if (city != null && !city.trim().isEmpty()) conditions.add("地址包含「" +
city + "」");
 if (email != null && !email.trim().isEmpty()) conditions.add("郵箱包含「" +
email + "」");
 if (minWeight != null && !minWeight.trim().isEmpty()) conditions.add("重量
≥ " + minWeight + "kg");
 if (maxWeight != null && !maxWeight.trim().isEmpty()) conditions.add("重量
≤ " + maxWeight + "kg");

 return conditions.isEmpty() ? "顯示所有顧客" : String.join("& ",
conditions);
 }

 private void performCustomValidation(Customer customer, BindingResult
bindingResult, Integer customerIndex) {
 // 檢查重複姓名 (編輯模式下排除自己)
 boolean nameExists = customerDatabase.stream()
 .anyMatch(c -> {
 if (customerIndex != null) {
 int currentIndex = customerDatabase.indexOf(c);
 return currentIndex != customerIndex.intValue() &&
 c.getName().equalsIgnoreCase(customer.getName());
 } else {
 return c.getName().equalsIgnoreCase(customer.getName());
 }
 });

 if (nameExists) {
 bindingResult.rejectValue("name", "duplicate.name", "姓名「" +
customer.getName() + "」已存在");
 }

 // 檢查重量範圍
 try {
 double weight = Double.parseDouble(customer.getWeight());
```



```
 if (weight < 20 || weight > 200) {
 bindingResult.rejectValue("weight", "invalid.weight.range", "重量
必須在 20-200 kg 之間");
 }
 } catch (NumberFormatException e) {
 bindingResult.rejectValue("weight", "invalid.weight.format", "重量格式
無效");
 }
}

private double calculateAverageWeight() {
 return customerDatabase.stream()
 .mapToDouble(Customer::getWeightAsDouble)
 .average()
 .orElse(0.0);
}

private List<Customer> getRecentCustomers(int limit) {
 return customerDatabase.stream()
 .sorted((a, b) -> b.getCreatedAt().compareTo(a.getCreatedAt()))
 .limit(limit)
 .collect(Collectors.toList());
}

private Map<String, Object> getSystemStats() {
 Map<String, Object> stats = new HashMap<>();
 stats.put("javaVersion", System.getProperty("java.version"));
 stats.put("osName", System.getProperty("os.name"));
 stats.put("totalMemory", Runtime.getRuntime().totalMemory() / (1024 *
1024) + " MB");
 stats.put("freeMemory", Runtime.getRuntime().freeMemory() / (1024 * 1024)
+ " MB");
 return stats;
}

private Map<String, Object> generateStatistics() {
 Map<String, Object> stats = new HashMap<>();

 stats.put("totalCustomers", customerDatabase.size());
 stats.put("averageWeight", calculateAverageWeight());
 stats.put("maxWeight",
customerDatabase.stream().mapToDouble(Customer::getWeightAsDouble).max().orElse(0)
);
 stats.put("minWeight",
customerDatabase.stream().mapToDouble(Customer::getWeightAsDouble).min().orElse(0)
);

 // 城市分佈
 Map<String, Long> cityDistribution = customerDatabase.stream()
 .collect(Collectors.groupingBy(c -> extractCity(c.getAddress()),
Collectors.counting()));
 stats.put("cityDistribution", cityDistribution);

 // 重量分佈
```

```
 long lightWeight = customerDatabase.stream().filter(c ->
c.getWeightAsDouble() < 60).count();
 long normalWeight = customerDatabase.stream().filter(c ->
c.getWeightAsDouble() >= 60 && c.getWeightAsDouble() <= 75).count();
 long heavyWeight = customerDatabase.stream().filter(c ->
c.getWeightAsDouble() > 75).count();

 Map<String, Long> weightDistribution = new HashMap<>();
 weightDistribution.put("輕量級", lightWeight);
 weightDistribution.put("標準級", normalWeight);
 weightDistribution.put("重量級", heavyWeight);
 stats.put("weightDistribution", weightDistribution);

 return stats;
 }

 private Map<String, Object> generateChartData() {
 Map<String, Object> chartData = new HashMap<>();

 // 重量趨勢數據
 List<Double> weights = customerDatabase.stream()
 .map(Customer::getWeightAsDouble)
 .collect(Collectors.toList());
 chartData.put("weights", weights);

 // 城市標籤
 Set<String> cities = customerDatabase.stream()
 .map(c -> extractCity(c.getAddress()))
 .collect(Collectors.toSet());
 chartData.put("cities", new ArrayList<>(cities));

 return chartData;
 }

 private String extractCity(String address) {
 if (address == null) return "未知";
 if (address.contains("台北")) return "台北";
 if (address.contains("新北")) return "新北";
 if (address.contains("桃園")) return "桃園";
 if (address.contains("台中")) return "台中";
 if (address.contains("台南")) return "台南";
 if (address.contains("高雄")) return "高雄";
 return "其他";
 }

 private List<String> getCityOptions() {
 return Arrays.asList("台北市", "新北市", "桃園市", "台中市", "台南市", "高雄
市", "其他");
 }
}
```

## 第五部分：進階 Thymeleaf 模板

### 🏠 homepage.html - 改良版首頁

```
<!DOCTYPE html>
<html lang="zh-TW" xmlns:th="http://www.thymeleaf.org">
<head>
 <meta charset="UTF-8"/>
 <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
 <title>Spring Boot Thymeleaf 顧客管理系統</title>
 <style>
 * { margin: 0; padding: 0; box-sizing: border-box; }
 body {
 font-family: 'Microsoft JhengHei', 'Segoe UI', Arial, sans-serif;
 line-height: 1.6;
 background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
 min-height: 100vh;
 }
 .container {
 max-width: 1200px;
 margin: 0 auto;
 padding: 20px;
 }
 .header {
 text-align: center;
 color: white;
 margin-bottom: 40px;
 padding: 40px 0;
 }
 .header h1 {
 font-size: 3em;
 margin-bottom: 10px;
 text-shadow: 2px 2px 4px rgba(0,0,0,0.3);
 }
 .header p {
 font-size: 1.2em;
 opacity: 0.9;
 }
 .dashboard {
 display: grid;
 grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
 gap: 20px;
 margin-bottom: 40px;
 }
 .card {
 background: white;
 border-radius: 15px;
 padding: 30px;
 box-shadow: 0 10px 30px rgba(0,0,0,0.2);
 transition: transform 0.3s, box-shadow 0.3s;
 }
 .card:hover {
```

```
 transform: translateY(-5px);
 box-shadow: 0 15px 40px rgba(0,0,0,0.3);
 }
 .card h3 {
 color: #2c3e50;
 margin-bottom: 15px;
 font-size: 1.4em;
 display: flex;
 align-items: center;
 gap: 10px;
 }
 .stats-grid {
 display: grid;
 grid-template-columns: repeat(2, 1fr);
 gap: 15px;
 margin-bottom: 20px;
 }
 .stat-item {
 background: #f8f9fa;
 padding: 15px;
 border-radius: 8px;
 text-align: center;
 }
 .stat-number {
 font-size: 2em;
 font-weight: bold;
 color: #3498db;
 display: block;
 }
 .stat-label {
 color: #7f8c8d;
 font-size: 0.9em;
 }
 .navigation-grid {
 display: grid;
 grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
 gap: 15px;
 }
 .nav-button {
 display: block;
 padding: 15px 20px;
 background: linear-gradient(135deg, #3498db, #2980b9);
 color: white;
 text-decoration: none;
 border-radius: 10px;
 text-align: center;
 font-weight: bold;
 transition: all 0.3s;
 border: none;
 cursor: pointer;
 }
 .nav-button:hover {
 background: linear-gradient(135deg, #2980b9, #1f4e79);
 transform: translateY(-2px);
 }
```

```
}
.nav-button.green {
 background: linear-gradient(135deg, #27ae60, #229954);
}
.nav-button.green:hover {
 background: linear-gradient(135deg, #229954, #1e7e34);
}
.nav-button.orange {
 background: linear-gradient(135deg, #f39c12, #e67e22);
}
.nav-button.orange:hover {
 background: linear-gradient(135deg, #e67e22, #d35400);
}
.nav-button.purple {
 background: linear-gradient(135deg, #9b59b6, #8e44ad);
}
.nav-button.purple:hover {
 background: linear-gradient(135deg, #8e44ad, #7d3c98);
}
.recent-customers {
 background: white;
 border-radius: 15px;
 padding: 30px;
 box-shadow: 0 10px 30px rgba(0,0,0,0.2);
 margin-top: 20px;
}
.customer-item {
 display: flex;
 justify-content: space-between;
 align-items: center;
 padding: 10px 0;
 border-bottom: 1px solid #ecf0f1;
}
.customer-item:last-child {
 border-bottom: none;
}
.customer-info strong {
 color: #2c3e50;
}
.customer-meta {
 font-size: 0.9em;
 color: #7f8c8d;
}
.system-info {
 background: #2c3e50;
 color: white;
 border-radius: 15px;
 padding: 20px;
 margin-top: 20px;
 font-size: 0.9em;
}
.system-info h4 {
 margin-bottom: 10px;
}
```

```

 .system-grid {
 display: grid;
 grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
 gap: 10px;
 }
 @media (max-width: 768px) {
 .stats-grid {
 grid-template-columns: 1fr;
 }
 .navigation-grid {
 grid-template-columns: 1fr;
 }
 }
 </style>
</head>
<body>
 <div class="container">
 <!-- 頁面標題 -->
 <div class="header">
 <h1>🏠 顧客管理系統</h1>
 <p>Spring Boot + Thymeleaf 完整解決方案</p>
 <p><small th:text="'當前時間：' + ${#dates.format(currentTime, 'yyyy年MM月dd日 HH:mm:ss')}">載入中...</small></p>
 </div>

 <!-- 儀表板卡片 -->
 <div class="dashboard">
 <!-- 統計摘要卡片 -->
 <div class="card">
 <h3>📊 系統統計</h3>
 <div class="stats-grid">
 <div class="stat-item">
 <span class="stat-number"
th:text="${customerCount}">0
 總顧客數
 </div>
 <div class="stat-item">
 <span class="stat-number"
th:text="${#numbers.formatDecimal(averageWeight, 1, 1)}">0
 平均重量(kg)
 </div>
 </div>
 </div>

 <!-- 功能導航卡片 -->
 <div class="card">
 <h3>🔗 快速操作</h3>
 <div class="navigation-grid">
 <a th:href="@{/customerlist}" class="nav-button">📋 顧客清單

 <a th:href="@{/customerForm}" class="nav-button green">➕ 新
增顧客
 <a th:href="@{/customerAdvancedSearch}" class="nav-button
orange">🔍 進階搜尋
 </div>
 </div>
 </div>
 </div>

```

```

 <a th:href="@{/customerStats}" class="nav-button purple">📊
統計報表
 </div>
</div>
</div>

<!-- 最近新增的顧客 -->
<div class="recent-customers" th:if="${recentCustomers != null and
#lists.size(recentCustomers) > 0}">
 <h3>🆕 最近新增的顧客</h3>
 <div th:each="customer : ${recentCustomers}" class="customer-item">
 <div class="customer-info">
 <strong th:text="${customer.name}">顧客姓名

 地址

 </div>
 <div class="customer-meta">
 重量

 <small th:text="${#dates.format(customer.createdAt, 'MM/dd
HH:mm')}">建立時間</small>
 </div>
 </div>
</div>

<!-- 系統資訊 -->
<div class="system-info">
 <h4>🖥️ 系統資訊</h4>
 <div class="system-grid">
 <div>Java 版本: Unknown</div>
 <div>作業系統: Unknown</div>
 <div>總記憶體: Unknown</div>
 <div>可用記憶體: Unknown</div>
 </div>
</div>
</body>
</html>

```

## 📁 第六部分：進階功能與最佳實務

### 🔍 customerform.html - 增強版表單

```

<!DOCTYPE html>
<html lang="zh-TW" xmlns:th="http://www.thymeleaf.org">
<head>
 <meta charset="UTF-8"/>

```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0"/>
<title th:text="${isEdit} ? '編輯顧客資料' : '新增顧客資料'">顧客表單</title>
<style>
 body {
 font-family: 'Microsoft JhengHei', Arial, sans-serif;
 margin: 0;
 background: linear-gradient(135deg, #a8edea 0%, #fed6e3 100%);
 min-height: 100vh;
 }
 .container {
 max-width: 600px;
 margin: 30px auto;
 background: white;
 padding: 40px;
 border-radius: 20px;
 box-shadow: 0 15px 35px rgba(0,0,0,0.1);
 }
 .form-header {
 text-align: center;
 margin-bottom: 30px;
 padding-bottom: 20px;
 border-bottom: 3px solid #e74c3c;
 }
 .form-header h1 {
 color: #2c3e50;
 margin: 0;
 font-size: 2.2em;
 }
 .form-header p {
 color: #7f8c8d;
 margin: 10px 0 0 0;
 }
 .alert {
 padding: 15px;
 margin-bottom: 20px;
 border-radius: 5px;
 border: 1px solid;
 }
 .alert-danger {
 background-color: #f8d7da;
 border-color: #f5c6cb;
 color: #721c24;
 }
 .alert-success {
 background-color: #d4edda;
 border-color: #c3e6cb;
 color: #155724;
 }
 .form-group {
 margin-bottom: 25px;
 position: relative;
 }
 .form-group label {
 display: block;
```



```
 margin-bottom: 8px;
 font-weight: bold;
 color: #34495e;
 font-size: 16px;
 }
 .form-group input, .form-group select {
 width: 100%;
 padding: 15px;
 border: 2px solid #bdc3c7;
 border-radius: 10px;
 font-size: 16px;
 transition: all 0.3s;
 box-sizing: border-box;
 background-color: #fff;
 }
 .form-group input:focus, .form-group select:focus {
 outline: none;
 border-color: #3498db;
 box-shadow: 0 0 10px rgba(52, 152, 219, 0.3);
 transform: translateY(-2px);
 }
 .form-group.has-error input {
 border-color: #e74c3c;
 background-color: #fdf2f2;
 }
 .error-message {
 color: #e74c3c;
 font-size: 14px;
 margin-top: 5px;
 display: block;
 }
 .required {
 color: #e74c3c;
 font-weight: bold;
 }
 .form-row {
 display: grid;
 grid-template-columns: 1fr 1fr;
 gap: 20px;
 }
 .form-actions {
 text-align: center;
 margin-top: 40px;
 padding-top: 30px;
 border-top: 2px solid #ecf0f1;
 }
 .btn {
 display: inline-block;
 padding: 15px 35px;
 margin: 0 10px;
 border: none;
 border-radius: 10px;
 font-size: 16px;
 font-weight: bold;
```

```

 text-decoration: none;
 cursor: pointer;
 transition: all 0.3s;
 min-width: 140px;
 }
 .btn-primary {
 background: linear-gradient(135deg, #27ae60, #229954);
 color: white;
 }
 .btn-primary:hover {
 background: linear-gradient(135deg, #229954, #1e7e34);
 transform: translateY(-3px);
 box-shadow: 0 10px 25px rgba(39, 174, 96, 0.4);
 }
 .btn-secondary {
 background: linear-gradient(135deg, #95a5a6, #7f8c8d);
 color: white;
 }
 .btn-secondary:hover {
 background: linear-gradient(135deg, #7f8c8d, #6c7b7d);
 transform: translateY(-3px);
 }
 .form-note {
 background: linear-gradient(135deg, #e8f5e8, #f0f8f0);
 padding: 20px;
 border-radius: 10px;
 border-left: 5px solid #27ae60;
 margin-bottom: 30px;
 color: #2d5a2d;
 }
 .character-counter {
 position: absolute;
 right: 10px;
 top: 45px;
 font-size: 12px;
 color: #7f8c8d;
 }
 @media (max-width: 768px) {
 .container {
 margin: 10px;
 padding: 20px;
 }
 .form-row {
 grid-template-columns: 1fr;
 gap: 10px;
 }
 .btn {
 display: block;
 margin: 10px 0;
 width: 100%;
 }
 }

```

</style>

</head>

```

<body>
 <div class="container">
 <!-- 表單標題 -->
 <div class="form-header">
 <h1 th:text="${isEdit} ? '✎ 編輯顧客資料' : '+ 新增顧客資料'">表單標題
 </h1>

 <p th:if="${isEdit}">修改現有顧客的詳細資料</p>
 <p th:unless="${isEdit}">輸入新顧客的完整資訊</p>
 </div>

 <!-- 錯誤訊息顯示 -->
 <div th:if="${validationErrors != null and #lists.size(validationErrors) >
0}" class="alert alert-danger">
 ⚠ 表單驗證失敗 :
 <ul style="margin: 10px 0 0 20px; padding: 0;">
 <li th:each="error : ${validationErrors}"
th:text="${error.defaultMessage}">錯誤訊息

 </div>

 <!-- 成功訊息 -->
 <div th:if="${successMessage != null}" class="alert alert-success"
th:text="${successMessage}">
 成功訊息
 </div>

 <!-- 表單提示 -->
 <div class="form-note">
 💡 填寫提示 :
 <ul style="margin: 10px 0 0 0; padding-left: 20px;">
 標有 * 的欄位為必填項目
 重量請以公斤為單位，支援小數點
 電子郵件將用於系統通知

 </div>

 <!-- 顧客表單 -->
 <form th:action="@{/customerSubmit}" th:object="${customer}" method="post"
onsubmit="return validateForm()">
 <!-- 隱藏的索引欄位（編輯模式） -->
 <input type="hidden" name="customerIndex" th:value="${customerIndex}"
th:if="${isEdit}"/>

 <!-- 基本資料 -->
 <div class="form-row">
 <div class="form-group"
th:classappend="${#fields.hasErrors('name')} ? 'has-error'">
 <label for="name">👤 顧客姓名 *
 </label>

 <input type="text"
 id="name"
 th:field="*{name}"
 required
 placeholder="請輸入完整姓名"

```

```

 maxLength="50"
 onkeyup="updateCharacterCounter('name', 50)"/>
 0/50
 <span th:if="${#fields.hasErrors('name')}"
 th:errors="*{name}"
 class="error-message">姓名錯誤
</div>

<div class="form-group"
th:classappend="${#fields.hasErrors('weight')} ? 'has-error'">
 <label for="weight">🏋️ 重量 (公斤) *</label>
 <input type="number"
 id="weight"
 th:field="*{weight}"
 required
 min="20"
 max="200"
 step="0.1"
 placeholder="例：65.5"
 onchange="calculateBMI()"/>
 <span th:if="${#fields.hasErrors('weight')}"
 th:errors="*{weight}"
 class="error-message">重量錯誤
</div>
</div>

<!-- 聯絡資訊 -->
<div class="form-row">
 <div class="form-group"
th:classappend="${#fields.hasErrors('email')} ? 'has-error'">
 <label for="email">✉️ 電子郵件</label>
 <input type="email"
 id="email"
 th:field="*{email}"
 placeholder="example@email.com"/>
 <span th:if="${#fields.hasErrors('email')}"
 th:errors="*{email}"
 class="error-message">郵件錯誤
 </div>

 <div class="form-group">
 <label for="phone">☎️ 電話號碼</label>
 <input type="tel"
 id="phone"
 th:field="*{phone}"
 placeholder="0912-345-678"
 pattern="[0-9]{4}-[0-9]{3}-[0-9]{3}"/>
 </div>
</div>

<!-- 地址資訊 -->
<div class="form-group"
th:classappend="${#fields.hasErrors('address')} ? 'has-error'">

```

```

 <label for="address">🏠 詳細地址 *
 </label>

 <input type="text"
 id="address"
 th:field="*{address}"
 required
 placeholder="請輸入完整地址"
 maxlength="200"
 onkeyup="updateCharacterCounter('address', 200)"/>
 0/200
 <span th:if="${#fields.hasErrors('address')}"
 th:errors="*{address}"
 class="error-message">地址錯誤
</div>

<!-- BMI 計算結果顯示 -->
<div class="form-group" style="background: #f8f9fa; padding: 15px;
border-radius: 8px;">
 <label>📊 BMI 計算 (假設身高 170cm)</label>
 <div id="bmi-result" style="font-size: 1.2em; color: #2c3e50;
font-weight: bold;">
 請輸入重量後自動計算
 </div>
</div>

<!-- 表單按鈕 -->
<div class="form-actions">
 <button type="submit" class="btn btn-primary">
 提交
 </button>
 <a th:href="@{/customerlist}" class="btn btn-secondary">✕ 取消

</div>
</form>

<!-- 回到首頁連結 -->
<div style="text-align: center; margin-top: 30px; padding-top: 20px;
border-top: 1px solid #ecf0f1;">
 <a th:href="@{/}" style="color: #3498db; text-decoration: none; font-
weight: bold;">🏠 回到首頁
</div>
</div>

<script>
 // 字符計數器更新
 function updateCharacterCounter(fieldId, maxLength) {
 const field = document.getElementById(fieldId);
 const counter = document.getElementById(fieldId + '-counter');
 const currentLength = field.value.length;

 counter.textContent = currentLength + '/' + maxLength;
 counter.style.color = currentLength > maxLength * 0.8 ? '#e74c3c' :
'#7f8c8d';
 }

```

```
}

// BMI 計算
function calculateBMI() {
 const weightField = document.getElementById('weight');
 const bmiResult = document.getElementById('bmi-result');

 const weight = parseFloat(weightField.value);
 const height = 1.7; // 假設身高 170cm

 if (weight && weight > 0) {
 const bmi = weight / (height * height);
 let category = '';
 let color = '';

 if (bmi < 18.5) {
 category = '過輕';
 color = '#3498db';
 } else if (bmi < 24) {
 category = '正常';
 color = '#27ae60';
 } else if (bmi < 27) {
 category = '過重';
 color = '#f39c12';
 } else {
 category = '肥胖';
 color = '#e74c3c';
 }

 bmiResult.innerHTML = `BMI: <span style="color:
${color}">${bmi.toFixed(1)} (${category})`;
 } else {
 bmiResult.textContent = '請輸入有效的重量';
 }
}

// 表單驗證
function validateForm() {
 const name = document.getElementById('name').value.trim();
 const weight = document.getElementById('weight').value;
 const address = document.getElementById('address').value.trim();

 if (!name) {
 alert('⚠ 請輸入顧客姓名');
 document.getElementById('name').focus();
 return false;
 }

 if (!weight || weight <= 0) {
 alert('⚠ 請輸入有效的重量');
 document.getElementById('weight').focus();
 return false;
 }
}
```

```
 if (!address) {
 alert('⚠️ 請輸入詳細地址');
 document.getElementById('address').focus();
 return false;
 }

 return true;
 }

 // 頁面載入時初始化
 document.addEventListener('DOMContentLoaded', function() {
 // 更新字符計數器
 updateCharacterCounter('name', 50);
 updateCharacterCounter('address', 200);

 // 計算初始BMI
 calculateBMI();

 // 電話號碼格式化
 document.getElementById('phone').addEventListener('input', function(e)
 {
 let value = e.target.value.replace(/\D/g, '');
 if (value.length >= 4) {
 value = value.substring(0,4) + '-' + value.substring(4);
 }
 if (value.length >= 9) {
 value = value.substring(0,9) + '-' + value.substring(9,12);
 }
 e.target.value = value;
 });
 });
</script>
</body>
</html>
```

---

## 📦 第七部分：學習總結與最佳實務

### 📖 核心知識回顧

#### 1. Spring Boot MVC 架構

- 控制器請求映射
- 模型資料傳遞
- 視圖解析機制

#### 2. Thymeleaf 模板引擎

- 表達式語言 (OGNL)
- 條件判斷和迴圈
- 表單資料綁定

### 3. 前端技術整合

- 響應式設計
- JavaScript 互動
- CSS 動畫效果

### 4. 進階功能實作

- 資料驗證機制
- 搜尋和過濾
- 統計圖表顯示

## 專業開發建議

#### 1. 效能最佳化

- 模板快取設定
- 靜態資源壓縮
- 資料庫查詢優化

#### 2. 安全性考量

- 輸入資料驗證
- XSS 攻擊防護
- CSRF 令牌保護

#### 3. 可維護性設計

- 程式碼模組化
- 註解和文件
- 單元測試覆蓋

## 進階學習方向

- **Spring Security** - 認證授權
- **Spring Data JPA** - 資料持久化
- **RESTful API** - 前後端分離
- **微服務架構** - Spring Cloud
- **容器化部署** - Docker/Kubernetes

這份 Spring Boot + Thymeleaf 完整教學涵蓋了從基礎到進階的所有重要概念，提供了實用的程式碼範例和最佳實務指引，是學習現代 Java Web 開發的寶貴資源！🔗